



MA 5104: Statistical Inference and Learning

Module 4: Learning

Lecture 28: Finishing the Linear Model, and Estimating Test Error

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An Example Before a Definition

A plant runs a process at two temperatures with two catalysts. Mean yield (%) in each of the four conditions:

	160°C	180°C
Catalyst A	60	78
Catalyst B	72	66

What is the effect of raising the temperature by 20°?

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An Example Before a Definition

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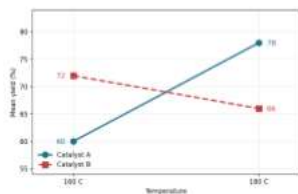
What is the effect of raising the temperature by 20°?

Averaging the two catalysts gives +12. That is correct arithmetic and useless advice: the true answer is +18 for catalyst A and -6 for catalyst B.

There is no such thing as “the effect of temperature” here. The question is malformed until you say *which catalyst*.

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Interaction: When the Lines Are Not Parallel



$$Y = T + C \quad \times$$

$$Y = T + C + T \cdot C \quad \checkmark$$

$\hat{\beta} \neq 0$
Active coefficient

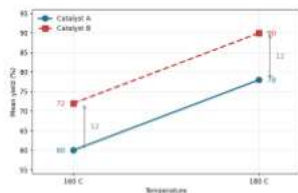
B beats A by 12 at 160° and loses by 12 at 180°. N

An **interaction** means the effect of one variable depends on the value of another. Parallel lines let you summarise each factor on its own; non-parallel lines do not, and **averaging will not rescue you**.

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The Additive Case: When Separate Summaries Work

Same experiment, one number changed: catalyst B at 180° now yields 90 instead of 66.



- Raising the temperature is worth **+18** — for **both** catalysts.
- Catalyst B beats A by **12** — at **both** temperatures.

Two sentences, each true without conditions.

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What "Linear Model" Actually Means

Linear in the parameters β_j , not necessarily in the original inputs. Sources of columns of X :

Transformations

$$X_j = \log x, \sqrt{x}, x^2$$

Categorical variables

$$X_j = 1(G = j)$$

Basis expansions

$$X_1 = x, X_2 = x^2, X_3 = x^3$$

Interactions

$$X_3 = X_1 \cdot X_2$$

Every one of these is a column of X .

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Categorical Predictors: Dummy Coding

A predictor with k levels enters as $k - 1$ indicator columns, with one level held out as the **reference**.

Three departments, reference = Maths:

$$\hat{y} = \beta_0 + \beta_1 1\{\text{Physics}\} + \beta_2 1\{\text{Chemistry}\} + \dots$$

so β_1 is the Physics–Maths difference, not a Physics "effect".

- Why $k - 1$ and not k : with all k columns plus an intercept, the columns sum to the intercept column. **X loses rank** and we are back on the previous failure slide.
- Changing the reference level changes every coefficient and **changes no**

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 + \epsilon$$

Catalyst	Temp	Y	C-A	C-B
A	160	.	1	0
A	180	.	1	0
B	160	.	0	1
B	180	.	0	1

$n=5$
 $p=3$

$$y = \beta_0 + \beta_1 \text{Temp} + \beta_2 \text{C-A} + \beta_3 \text{C-B} + \epsilon$$

$$X = \begin{bmatrix} 1 & 160 & 1 & 0 \\ 1 & 180 & 1 & 0 \\ 1 & 160 & 0 & 1 \\ 1 & 180 & 0 & 1 \end{bmatrix}$$

D_1 D_2 $C-A$ $C-B$ 5×4

$$\begin{bmatrix} y_2 - y_1 \\ \vdots \\ y_2 - y_1 \end{bmatrix}$$

- why $k = 1$ and not k . with an k columns plus an intercept, the columns sum to the intercept column. X loses rank and we are back on the previous failure slide.
- Changing the reference level changes every coefficient and **changes no fitted value whatsoever**.
- Ordinal predictors: coding as 1, 2, 3 asserts equal spacing. That is an assumption, and usually a wrong one.

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Write the correct model between

Education - 10th, 12th, BSc, MSc, PhD
 Occupation-types - Blue collar, white collar

$$\text{income } (Y) = f(\text{education } (X_1), \text{occupation_types } (X_2))$$

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The common slope model-(assuming no interactions!)

The regression model for income (Y) v.s. (education X and occupation types D_1)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \varepsilon_i, \quad i = 1, \dots, n.$$

- For Blue collar group ($D_1 = 0$):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

- For White collar group ($D_1 = 1$):

$$Y_i = (\beta_0 + \gamma_1) + \beta_1 X_i + \varepsilon_i,$$

- Here, both groups have their own regression lines. Same slope but different intercept. It means that they are **parallel!**

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The interaction model

The regression model for income (Y) v.s. (education X and occupation types D_1)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \delta_1 (X_i D_{1i}) + \varepsilon_i, \quad i = 1, \dots, n.$$

- For Blue collar group ($D_1 = 0$):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

- For White collar group ($D_1 = 1$):

$$Y_i = (\beta_0 + \gamma_1) + (\beta_1 + \delta_1) X_i + \varepsilon_i,$$

- Each group has different slope and different intercept. If there is interaction, they are **NOT parallel!**.
- In other words, the value of the categorical predictor also affects the association between Y (income) and X (education);

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$$\begin{matrix} & \begin{matrix} 1 & 180 & 0 \\ D_1 & D_2 & C-N \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{matrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{matrix} \end{matrix} \quad \begin{matrix} 1 & -1 \\ 2 & 0 \\ 3 & 1 \end{matrix} \quad \begin{matrix} 5 \times 4 \\ 1 \\ 0 \\ 1 \end{matrix}$$

$$\begin{pmatrix} Y_2 - Y_1 \\ Y_3 - Y_2 \end{pmatrix}$$

$$y = \beta_0 + \beta_1 \text{Temp} + \beta_2 C-N + \varepsilon$$

$$\begin{aligned} y_t &= \hat{\beta}_0 + \hat{\beta}_1 t \\ y_{t+1} &= \hat{\beta}_0 + \hat{\beta}_1 (t+1) \end{aligned}$$

$$\hat{\beta}_1 = y_{t+1} - y_t$$

Given Temp,
 if C-N by 1 unit
 then $\hat{\beta}_2 \rightarrow$ change in y

Reading the Output: A Worked Case

Starting salary (Rs. lakh p.a.) on CGPA and programme, reference = M.Sc.:

$$\widehat{\text{salary}} = 4.20 + 1.35 \text{ CGPA} + 2.10 \cdot \mathbf{1}\{\text{M.Tech}\} + 0.45 \text{ CGPA} \cdot \mathbf{1}\{\text{M.Tech}\}$$

Coef.	Reads as	Meaning
4.20	intercept	

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4.20	intercept	M.Sc. student with CGPA = 0 . Nobody. An extrapolation.
1.35	slope	

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4.20	intercept	M.Sc. student with CGPA = 0 . Nobody. An extrapolation.
1.35	slope	per CGPA point, <i>for M.Sc.</i>
2.10	dummy	

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2.10	dummy	M.Tech salary at CGPA = 0 .
0.45	interaction	

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4.20	intercept	M.Sc. student with CGPA = 0 . Nobody. An extrapolation.
1.35	slope	per CGPA point, <i>for M.Sc.</i>
2.10	dummy	M.Tech salary at CGPA = 0 .
0.45	interaction	extra rise per CGPA point for M.Tech; their slope is 1.80.

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Cross-Validation

Validation Set

Split once: fit on one part, score on the other.

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Validation Set

Split once: fit on one part, score on the other.

Honest, but wasteful (you fit on half the data) and

highly variable since a different split gives a different answer.

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K-Fold

Split into K parts. Fit K times, each time holding out one part and scoring on it. Average the K errors.



- Every observation is used for scoring exactly once, and for fitting $K - 1$ times.
- K has its own bias-variance trade: small K means each fit uses less data, so test error is over-estimated (bias); large K means the K training sets overlap heavily, so the K errors are correlated and the average is noisier. $K = 5$ or 10 is the usual compromise.

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